

Размеры тензоров и FLOPs

S - размер батча B B -same

Вход: $X \in \mathbb{R}^{B \times 3 \times S \times S}$ $S: 16$

~~1~~ $1 \text{ MAC} = 2 \text{ FLOPs}$ и $\text{FP32} = 4 \text{ FLOPs}$

Размеры | Тензор

$B \times 3 \times S \times S$ Input

$B \times 32 \times S/2 \times S/2$ CONV1

$B \times 32 \times S/4 \times S/4$ MaxPool

$B \times 64 \times S/4 \times S/4$ CONV2

$B \times 128 \times S/8 \times S/8$ CONV3

$B \times 256 \times S/8 \times S/8$ CONV4

$B \times 256 \times S/16 \times S/16$ CONV5

$B \times 512 \times S/16 \times S/16$ CONV6

$B \times 512$ GAP

$B \times 256$ FC1

$B \times 100$ FC2

FLOPs чётко: $C_{in} \cdot k^2 \cdot \text{MAC} \Rightarrow$

$\Rightarrow 2 \cdot B \cdot C_{out} \cdot H_{out} \cdot W_{out} \cdot C_{in} \cdot k^2$

$F_1 = 2 \cdot B \cdot 32 \cdot (S/2)^2 \cdot 3 \cdot 7^2 = 2352 B S^2$

$F_2 = 6400 B S^2$; $F_3 = 2304 B S^2$

$F_4 = 1024 B S^2$; $F_5 = 4608 B S^2$

$F_6 = 1024 B S^2$

$F_{FC1} = 2 \cdot B \cdot 612 \cdot 256 = 262144 B$

$F_{FC2} = 51200 B$

$$\Rightarrow FLOPs(S, B) = 12412BS^2 + 313344B$$

FLOPS for RELU, MaxPool, Global
AvgPool не учитываемся - не влияют на сложность

Анализировать можно так

memory ≤ 9 , (param + input + act)

$$\text{input} = 3BS^2; \text{Conv1} = 3 \cdot 2B(S/2)^2 = 8BS^2$$

$$\text{MaxPool} = 2BS^2; \text{Conv2} = 4BS^2; 4BS^2$$

$$\text{Conv3} = \cancel{2BS^2}, \text{Conv4} = \cancel{2BS^2}$$

$$\text{Conv5} = BS^2; \text{Conv6} = 2BS^2$$

суммарно по параметрам:

$$(3 + 8 + 2 + 4 + 2 + 4 + 1 + 2)BS^2 = 26BS^2$$

$$\text{kernel: } B(512 + 256 + 100) = 868B$$

Параметры Conv:

$$P_{\text{conv}} = 3 \cdot 32 \cdot 7^2 + 32 \cdot 64 \cdot 5^2 + 64 \cdot 128 \cdot 3^2 + 128 \cdot 256 \cdot 3^2 + 256 \cdot 256 \cdot 3^2 + 256 \cdot 512 = 883296$$

$$\text{Linear: } 512 \cdot 256 + 256 + 256 \cdot 100 + 100 = 157028$$

$$\text{Всего } P = 1040324$$

$$\text{Memory}(S, B) = 104BS^2 + 3472B + 4161296$$

RELU inplace=True $\Rightarrow$ не годится.

Bytes moved
worst-case des Komputations
u. Datentransfers von.

$$D_{conv} = 4(B C_{in} H_{in} W_{in} + C_{in} C_{out}^2 + B C_{out} H_{out} W_{out})$$

$$D_{inplace} = 4(B C_{in} + C_{in} C_{out} + C_{out} + B C_{out})$$

$$D_{relu} = 4B(N+1) = 8BN$$

$$D_{op} = 4B(N_{in} + N_{out})$$

$$\text{Bytes Moved}(S, B) = 354 BS^2 + 859B + 4151296$$

Latency u. Energy

$$\Theta_L = (t_0, R_F, R_M)$$

3000000
ops/sec

3000
Mansb. FLOPs

3000000
max. mem. use.
max. mem. byte/s

$$T_{compute} = \frac{FLOPs(S, B)}{R_F}$$

$$T_{memory} = \frac{\text{Bytes Moved}(S, B)}{R_M}$$

$$\text{Latency}(S, B, \Theta_L) = t_0 + \max\left(\frac{FLOPs(S, B)}{R_F}, \frac{\text{Bytes Moved}(S, B)}{R_M}\right)$$

$$= t_0 + \max\left(\frac{B(127125^2 + 313344)}{R_F}, \frac{354BS^2 + 859B + 4151296}{R_M}\right)$$

$$+ 8532B + 41629C)$$

North Carolina

$$t_0 \approx 47,3 \text{ } \mu\text{S}$$

$$R_f \approx 14,2 \text{ TFLOPS}$$

$$R_M \approx 296 \text{ CB/s}$$

Исправно

$$\Theta \in (P_0, e_r, e_m)$$

Энергия
на GB

перелом
данной

 $\frac{1}{6}$

das ist die
Magnetkraft B_{LW}

Примечание

1/0 FLO

STOP

$$E(S, B, \theta_{\frac{\pi}{2}}) = p_0 \cdot \text{Latency}(S, B)^4$$

$$+ e_f \frac{\text{FLOPs}(S, B)}{10^9} + e_m \frac{\text{Bytes Moved}(S, B)}{10^9}$$

fitted bolometer: $P_0 \approx 1.34 \cdot 10^{-10} \text{ W}$

$$L_A \cong 0,01993 \text{ J/GFLIP}$$

$$e_m \approx 5,2 \cdot 10^{-14} \text{ J/GB}$$

FLOPs, bytes moved a latency U_{lat} to
коррелирует, поэтому канонизация
абстрагировать через FLOPs